

Flash-Crash Risk Loop

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How automation amplifies shocks

PROFESSIONAL DEFINITION

The Flash-Crash Risk Loop describes how automated trading can amplify shocks through feedback — rapid selling triggering further automated selling — producing sudden, severe market dislocations.

WHO DEVELOPED IT

Analysed in studies of events such as the 2010 Flash Crash and broader research on automation and market stability.

WHY IT WAS DEVELOPED

It was developed to explain how speed and automation can interact to destabilise markets, informing safeguards.

FIGURE 1 · FLASH-CRASH RISK LOOP

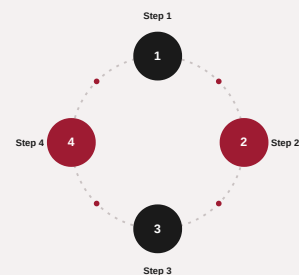


Figure 1. A schematic of the Flash-Crash Risk Loop as applied in BARBM312 (illustrative).

HOW IT IS USED

Feedback dynamics are mapped to identify amplification risks; circuit breakers and controls are designed to interrupt the loop.

WHEN IT IS USED

When assessing systemic risk in automated markets and designing safeguards.

BY WHOM IT IS USED

Regulators, risk managers and trading-system designers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It frames automation as a potential amplifier of shocks, motivating controls such as circuit breakers.

RELATED FRAMEWORKS IN THIS COURSE

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Latency Arms Race · Slibrary 24

Market Microstructure Map · Slibrary 24

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SOURCES (HARVARD REFERENCING STYLE)

Kirilenko, A. et al. (2017) 'The Flash Crash: High-frequency trading in an electronic market', Journal of Finance, 72(3).
CFTC-SEC (2010) Findings Regarding the Market Events of May 6, 2010.